

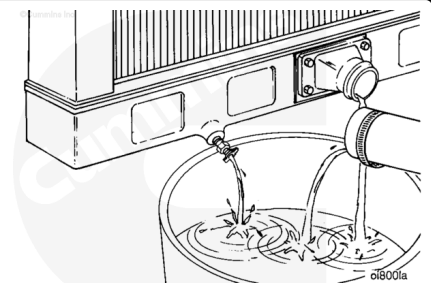
Trainings ▾

Preparatory Steps

Industrial and Power Generation

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

The turbocharger and turbocharger components can remain hot after the engine has been shut down or secured. To reduce the possibility of fire, property damage, burns, or other serious personal injury, allow the turbocharger system to cool down before beginning this procedure or repair and make sure that no combustible materials are located where it might come in contact with a hot turbocharger or turbocharger components.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

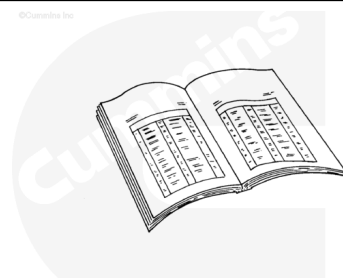
- Disconnect the batteries. See equipment manufacturer service information.

- Drain the cooling system to a point below the turbochargers. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)
- Disconnect the turbocharger coolant supply and return lines. Refer to Procedure 010-041 in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-041.html)
- Disconnect the turbocharger oil supply lines. Refer to Procedure 010-046 in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-046-tr.html)
- Disconnect the turbocharger oil drain lines. Refer to Procedure 010-045 in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-045-tr.html)
- Remove the required exhaust connection pipework. Refer to Procedure 011-001 in Section 11.
(/qs3/pubsys2/xml/en/procedures/56/56-011-001-tr.html) Refer to Procedure 011-002 in Section 11.
(/qs3/pubsys2/xml/en/procedures/56/56-011-002.html)

QSK60 Marine Applications

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



ck800wa

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

The turbocharger and turbocharger components can remain hot after the engine has been shut down or secured. To reduce the possibility of fire, property damage, burns, or other serious personal injury, allow the turbocharger system to cool down before beginning this procedure or repair and

make sure that no combustible materials are located where it might come in contact with a hot turbocharger or turbocharger components.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. See equipment manufacturer service information.
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html>)
- Remove the aftercooler heat shield panels.
- Remove the aftercooler heat shields. Refer to Procedure 010-129 in Section 10. (</qs3/pubsys2/xml/en/procedures/56/56-010-129-tr.html>)
- Remove the turbocharger heat shields. Refer to Procedure 010-076 in Section 10. (</qs3/pubsys2/xml/en/procedures/56/56-010-076-tr.html>)
- Remove the turbocharger inlet connections. Refer to Procedure 010-130 in Section 10. (</qs3/pubsys2/xml/en/procedures/56/56-010-130-tr.html>)
- Disconnect the turbocharger coolant supply and return lines. Refer to Procedure 010-041 in Section 10. (</qs3/pubsys2/xml/en/procedures/56/56-010-041.html>)
- Disconnect the turbocharger oil supply lines. Refer to Procedure 010-046 in Section 10. (</qs3/pubsys2/xml/en/procedures/56/56-010-046-tr.html>)
- Disconnect the turbocharger oil drain lines. Refer to Procedure 010-045 in Section 10. (</qs3/pubsys2/xml/en/procedures/56/56-010-045-tr.html>)
- Remove the required exhaust connection pipework. Refer to Procedure 011-001 in Section 11. (</qs3/pubsys2/xml/en/procedures/56/56-011-001->

tr.html) Refer to Procedure 011-002 in Section 11.
(/qs3/pubsys2/xml/en/procedures/56/56-011-002.html)

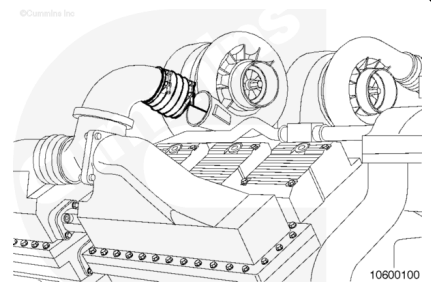
Remove

QSK45 Single Stage

Loosen the hose clamps on the hose between the air crossover and turbocharger.

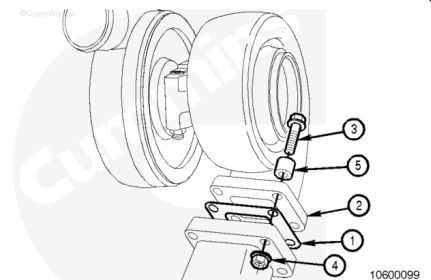
Slide the hose onto the air crossover.

Note : Do not disconnect the air crossover from the aftercooler.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Note the position of the spacers and direction of the capscrews reassembly.

Remove the four capscrews (3), spacers (5), and flange nuts (4).

Remove the turbocharger (2) and gasket (1).

Discard the gasket (1), capscrews (3), and flange nuts (4).

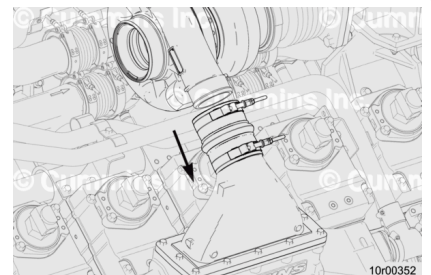
QSK60 Single Stage Industrial and Power Generation

Note : The QSK60 Single Stage has each high-pressure turbocharger supplying its own aftercooler cover.

Loosen the hose clamps on the hose between the air crossover and turbocharger.

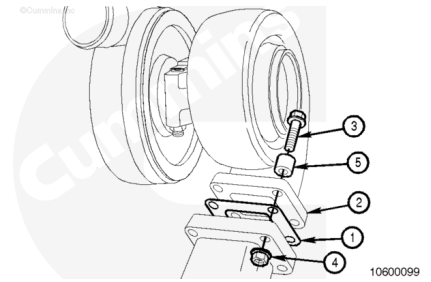
Slide the hose onto the aftercooler inlet connection.

Note : Do not disconnect the air crossover from the aftercooler.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Note the position of the spacers and direction of the capscrews reassembly.

Remove the four capscrews (3), spacers (5), and flange nuts (4).

Remove the turbocharger (2) and gasket (1).

Discard the gasket (1), capscrews (3), and flange nuts (4).

Cover openings. Use protective caps from Air Handling Clean Care Kit, Part Number 5394292.

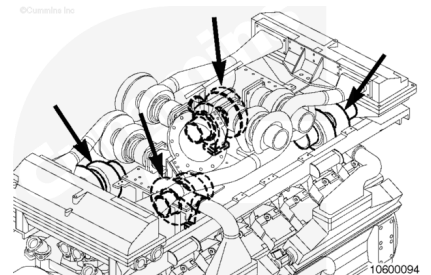
QSK60 Two Stage Industrial and Power Generation

High-pressure turbochargers are used on all QSK45 and QSK60 engines and are mounted directly to the exhaust manifold. The QSK60 two-stage engine also has four low-pressure turbochargers.

Removal of a high-pressure turbocharger on a QSK60 two-stage engine often requires the removal of low-pressure turbochargers and/or other components.

Determine which of the four high-pressure turbochargers to remove.

- Left bank rear
- Right bank front
- Left bank front
- Right bank rear.



The left bank rear and right bank front turbochargers are located under the low-pressure turbochargers. Remove the low-pressure turbocharger directly above the high-pressure turbocharger that requires removal. Refer to Procedure 010-035 in Section 10. (</qs3/pubsys2/xml/en/procedures/56/56-010-035-tr.html>)

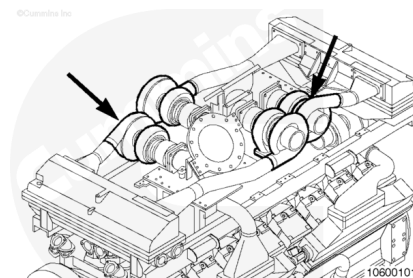
The left bank front and right bank rear turbochargers are located under the exhaust outlet connection and require removal of the exhaust outlet and mounting base, as detailed in this procedure.

Left Bank Rear or Right Bank Front

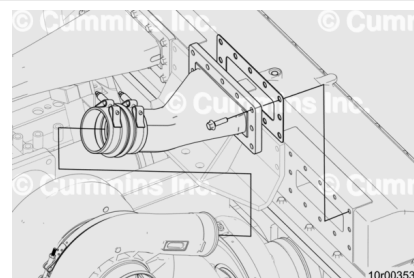
The left bank rear and right bank front turbochargers are located under the low-pressure turbochargers.

Remove the low-pressure turbocharger directly above the high-pressure turbocharger requiring removal. Refer to Procedure 010-035 in Section 10.

(/qs3/pubsys2/xml/en/procedures/56/56-010-035-tr.html)

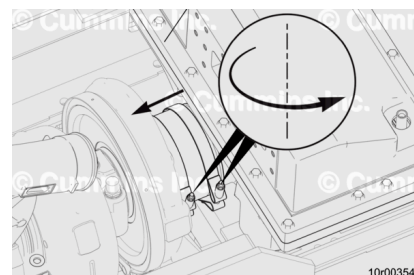


Remove the intercooler intake air crossover, located between the low-pressure turbocharger and the intercooler, to the intercooler.

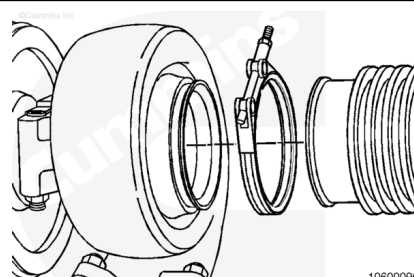


Loosen the two hose clamps on the hose between the turbocharger inlet and the intercooler.

Slide the hose toward the turbocharger.



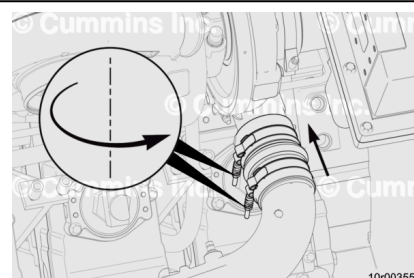
Remove the V-band clamp at the bellows to the turbine side of the turbocharger.



Loosen the hose clamps on the hose between the air crossover and turbocharger.

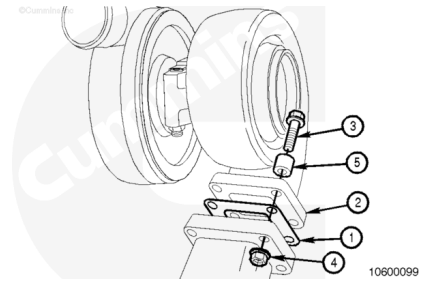
Slide the hose down the turbocharger outlet.

Note : Do not disconnect the air crossover from the aftercooler.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Note the position of the spacers and direction of the cap screws reassembly.

Remove the four cap screws (3), spacers (5), and flange nuts (4).

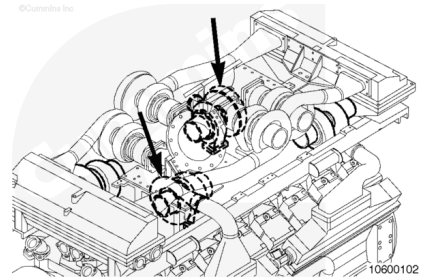
Remove the turbocharger (2) and gasket (1).

Discard the gasket (1), cap screws (3), and flange nuts (4).

Left Bank Front or Right Bank Rear

The left bank front and right bank rear turbochargers are located under the exhaust outlet connection and require removal of the exhaust outlet and mounting base.

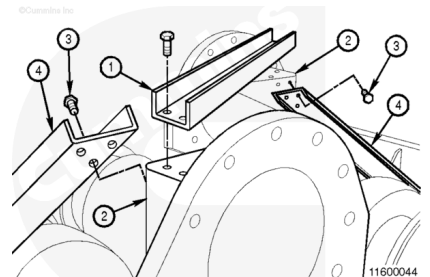
Remove the exhaust outlet and mounting base.



Remove the four cap screws from each end of the top cross support brace (1) between the two exhaust outlet connections (2).

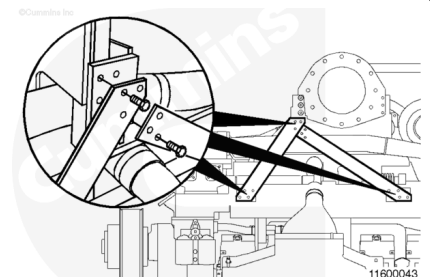
Remove the top cross support brace (1).

Remove the three cap screws (3) from each end of the two inboard support struts (4).



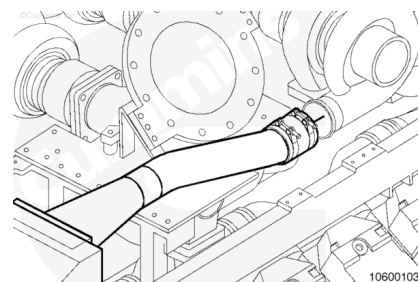
Remove the seven cap screws from each of the support braces between the exhaust outlet mounting base and the two-stage frame.

Remove the support braces.



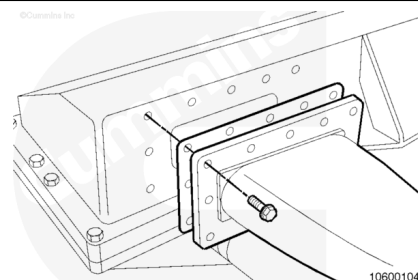
Loosen the clamps on the hose between the low-pressure turbocharger and the intercooler intake crossover.

Slide the hose and clamps onto the air crossover.

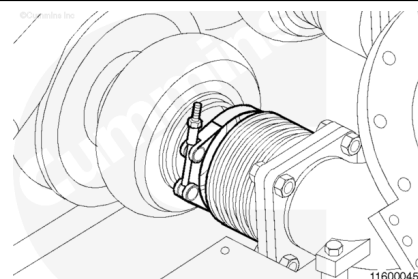


Remove the twelve capscrews connecting the intercooler intake crossover to the intercooler.

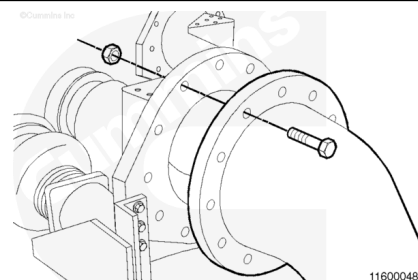
Remove and discard the gasket.



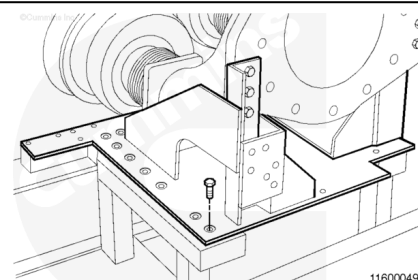
Disconnect, at the turbocharger, the V-band clamps connecting the bellows on the legs of the exhaust outlet flange to the low-pressure turbochargers.



Disconnect the exhaust piping from the exhaust outlet flange.

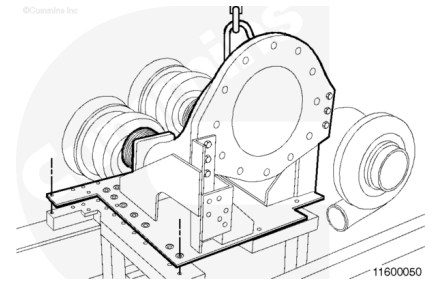


Remove the thirty capscrews connecting the exhaust outlet flange mounting base to the two-stage frame.



⚠ WARNING ⚠

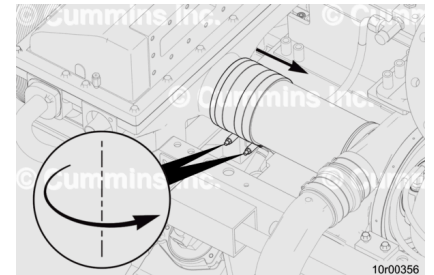
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



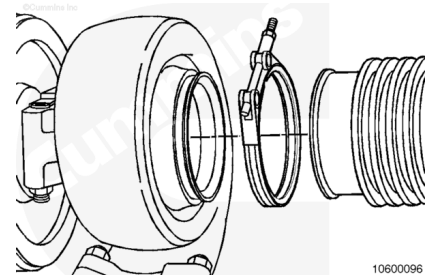
Remove the exhaust outlet flange and mounting base from the two-stage frame.

Loosen the two hose clamps on the hose between the turbocharger inlet and the intercooler.

Slide the hose toward the turbocharger.



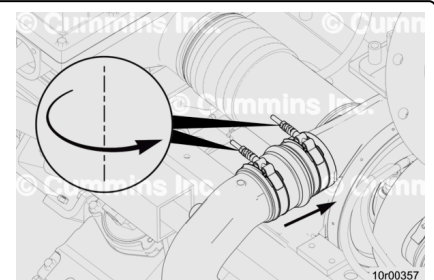
Remove the V-band clamp at the bellows to the turbine side of the turbocharger.



Loosen the hose clamps on the hose between the air crossover and turbocharger.

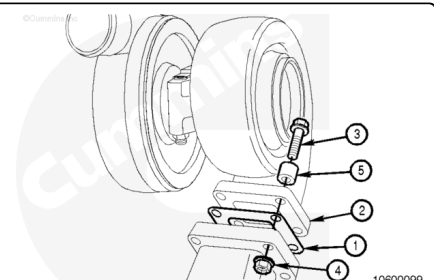
Slide the hose down the turbocharger outlet.

Note : Do not disconnect the air crossover from the aftercooler.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Note the position of the spacers and direction of the capscrews reassembly.

Remove the four capscrews (3), spacers (5), and flange nuts (4).

Remove the turbocharger (2) and gasket (1).

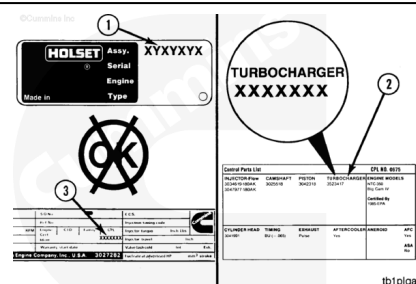
Discard the gasket (1), capscrews (3), and flange nuts (4).

Install

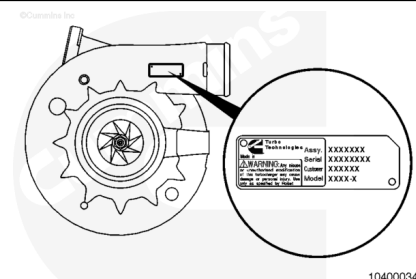
Check for Correct Component

Compare the assembly number (1) on the turbocharger dataplate with the turbocharger specified in the Control Parts List (CPL), Bulletin 3379133 (2).

The CPL number for each engine is listed on the engine's dataplate (3).



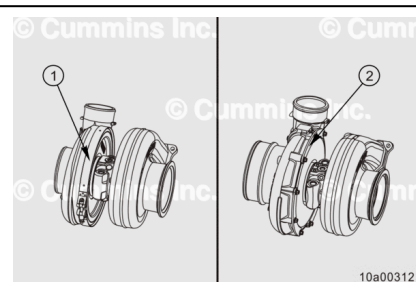
Holset® turbochargers are used on the QSK45 and QSK60 engines. Use the data tag on the turbocharger to determine the model.



The HE850 model name is unable to be retroactively applied to early design releases due to regulatory compliance constraints. These HE850 turbochargers will still have HE800 as the model name listed on the turbocharger dataplate. For differentiation from the standard HE800 turbocharger, the HE850 model name will be referenced throughout this procedure.

The procedure and specifications will be different based on the type of turbocharger. Identify the type of turbocharger equipped:

- The HE800 turbocharger (1) can be identified by the V-band joint between the compressor and bearing housing.
- The HE850 turbocharger (2) can be identified by the bolted joint between the compressor and bearing housing.



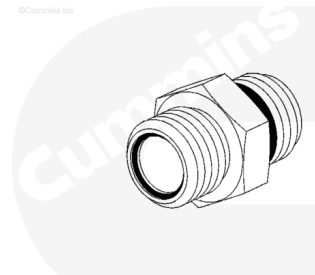
One turbocharger is mounted directly on the exhaust manifold.

Both the HE800 turbocharger and HE850 turbocharger are cooled using coolant from the high-temperature circuit.

Each HE800 turbocharger weighs approximately 57 kg [126 lb]. Each HE850 turbocharger weighs approximately 65 kg [143 lb]. Both will require proper lifting equipment.

Note : All turbocharger oil and water fittings are the flat-face o-ring types of fittings. Make sure the o-ring is in place before attaching the hose(s) to the fittings.

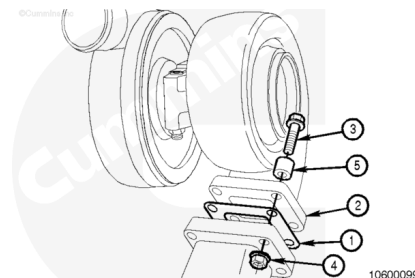
The Holset® turbocharger uses a 9/16-18 UNF, straight-thread, o-ring type of oil supply fitting.



QSK45 Single Stage

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



⚠ CAUTION ⚠

Do not reuse old capscrews (3) or flange nuts (4). Discard these components. Reusing these components can contribute to exhaust leaks.

Note : Make sure that the turbocharger is correctly aligned during installation to prevent damage to the turbocharger bellows.

Apply an anti-seize compound to the mounting capscrews threads.

The word OUT or the raised gasket bead **must** be positioned toward the turbocharger.

Install the gasket.

Do **not** install the turbocharger without spacers.

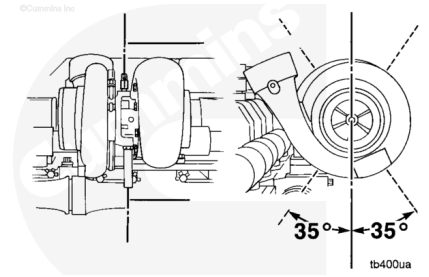
Install the turbocharger (2) and gasket (1) with four capscrews (3), spacers (5), and flange nuts (4).

Tighten the flange nuts (4).

Torque Value: 55 n•m [40 ft-lb]

The turbocharger uses capscrews to attach the bearing housing to the turbine housing.

Position the turbocharger drain tube. The drain tube **must** be within 35 degrees of vertical. Turn the bearing housing to align the tube, if necessary.

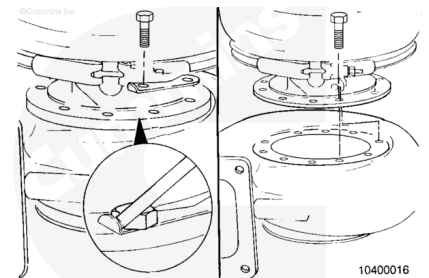


Bearing Housing Adjustment

Bend the lockplate off the capscrew heads.

Remove the capscrews to allow the bearing housing to rotate.

Lift the compressor and bearing housing. Rotate the bearing housing. Align the drain tube and the capscrew holes.



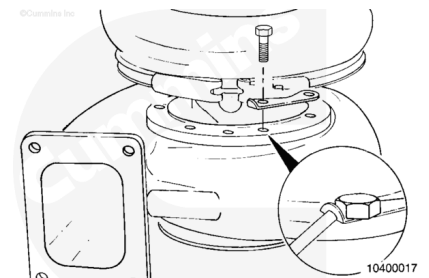
Apply anti-seize compound, Part Number 3824879, to the capscrews.

Install the lockplates and capscrews.

Tighten the capscrews.

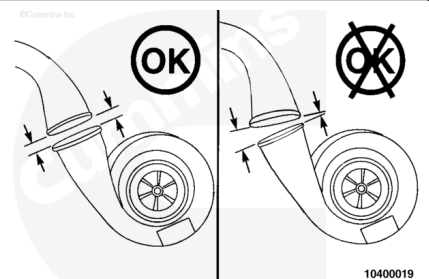
Torque Value: 21 n•m [186 in-lb]

Bend the lockplate tabs over the capscrews.



Check the air crossover-to-turbocharger alignment.

- Install the air crossover on the aftercooler.
- Look at the alignment between the crossover and the turbocharger compressor outlet.
- Remove the crossover and proceed with the assembly if the alignment is correct.
- Proceed to the next step to adjust the compressor housing if the alignment is **not** correct.

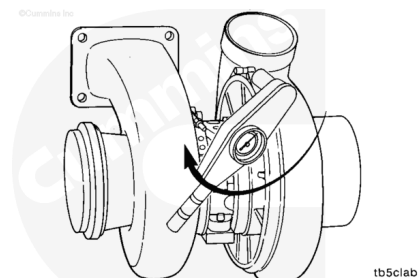


To adjust the turbocharger compressor housing alignment:

- Loosen the V-band clamp.
- Turn the compressor housing to the correct alignment.
- Tighten the clamp.

Torque Value: 9 n•m [80 in-lb]

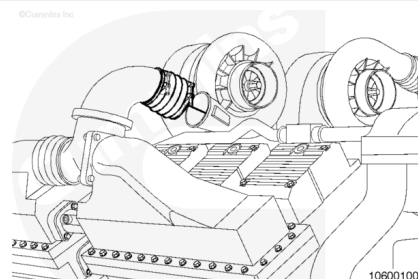
- Tap the clamp with a mallet. Tighten the clamp again.



Install the hose and two hose clamps between the turbocharger and the aftercooler inlet connection.

Tighten the clamps.

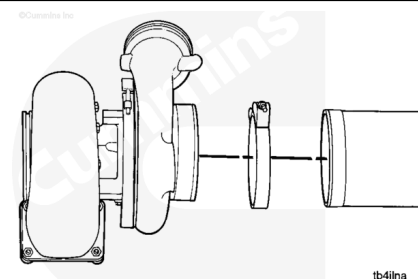
Torque Value: 9 n•m [80 in-lb]



Connect the inlet piping to the turbocharger.

Tighten the clamps.

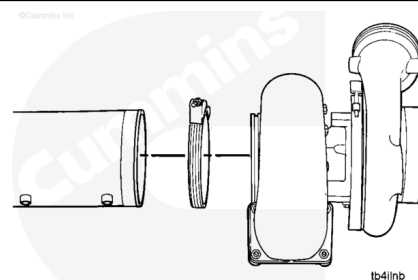
Torque Value: 9 n•m [80 in-lb]



Connect the exhaust piping to the turbocharger.

Tighten the V-band clamp.

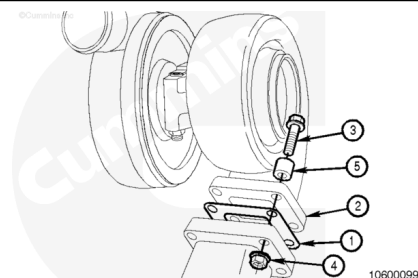
Torque Value: 9 n•m [80 in-lb]



QSK60 Single Stage Industrial and Power Generation

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



⚠ CAUTION ⚠

Do not reuse old capscrews (3) or flange nuts (4). Discard these components. Reusing these components can contribute to exhaust leaks.

⚠ CAUTION ⚠

Check to be sure the correct turbocharger assembly is installed. The turbocharger assembly number is on the dataplate that is mounted on the turbocharger. Compare the assembly number with the number in the Control Parts List (CPL) manual with the corresponding CPL for the engine being serviced. Many turbochargers appear the same physically, but contain different parts. If the wrong turbocharger assembly is installed, the performance of the engine will be less than specification, or damage can result to the engine due to excessive cylinder pressure and temperature.

Note : Replacement turbochargers are often built with a standard turbine housing and compressor housing orientation. These housing orientations on the replacement turbocharger may not match the housing orientations of the turbocharger that is being replaced. Make sure that the turbocharger is correctly aligned during installation to prevent damage to the turbocharger bellows.

Remove any protective caps and tape used to cover turbine inlet and exhaust manifold outlet.

Apply film of copper-free, high temperature anti-seize compound, Part Number 3824879, or equivalent, to turbocharger mounting capscrews.

The word OUT or the raised gasket bead **must** be positioned toward the turbocharger.

Install the gasket.

Do **not** install the turbocharger without spacers.

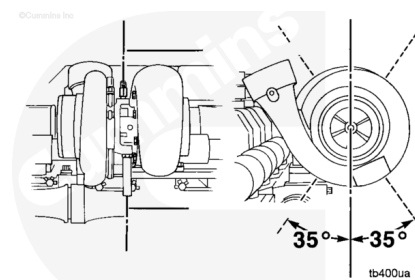
Install the turbocharger (2) and gasket (1) with four capscrews (3), spacers (5), and flange nuts (4).

Tighten the flange nuts (4).

Torque Value: 55 n•m [40 ft-lb]

The turbocharger uses capscrews to attach the bearing housing to the turbine housing.

Position the turbocharger drain tube. The drain tube **must** be within 35 degrees of vertical. Turn the bearing housing to align the tube, if necessary.

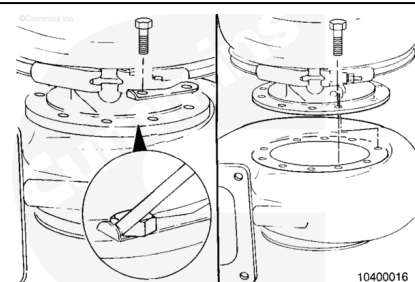


Bearing Housing Adjustment

Bend the lockplate off the capscrew heads.

Remove the capscrews to allow the bearing housing to rotate.

Lift the compressor and bearing housing. Rotate the bearing housing. Align the drain tube and the capscrew holes.



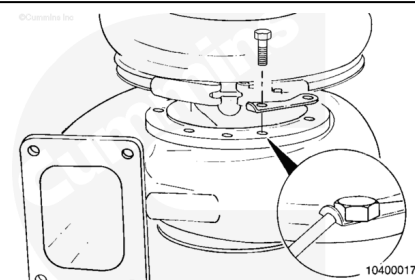
Apply anti-seize compound, Part Number 3824879, to the capscrews.

Install the lockplates and capscrews.

Tighten the capscrews.

Torque Value: 21 n•m [186 in-lb]

Bend the lockplate tabs over the capscrews.



High-Pressure Turbocharger:

The HE850 turbocharger does **not** use lockplate tabs.

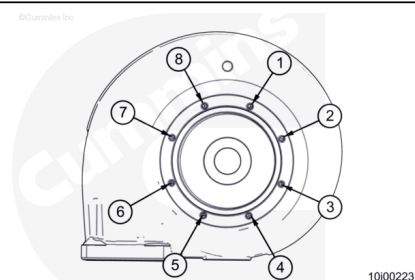
If turbine housing orientation with respect to bearing housing does **not** match turbocharger being replaced, place turbocharger on turbine housing outlet flange on clean sturdy work bench.

Secure the turbocharger and remove turbine housing to bearing housing mounting capscrews (3).

Rotate turbine housing to match orientation of turbocharger being replaced.

Install turbine housing to bearing housing mounting capscrews.

When torquing the turbine housing to bearing housing mounting capscrews, use the torque values and order of sequence shown below.

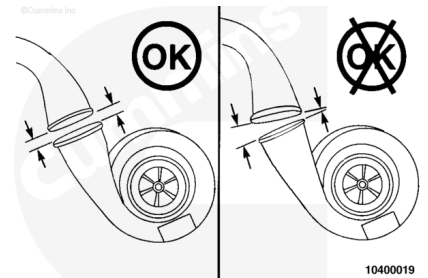


Turbine housing bolts shall be double tightened. Tighten bolts once in the circular sequence shown and repeat tightening each bolt again to the same value in the same circular sequence.

Torque Value: 40 n•m [30 ft-lb]

Check the air crossover-to-turbocharger alignment.

- Install the air crossovers on the aftercooler.
- Look at the alignment between the crossover and the turbocharger compressor outlet.
- Remove the crossover and proceed with the assembly if the alignment is correct.
- Proceed to the next step to adjust the compressor housing if the alignment is **not** correct.

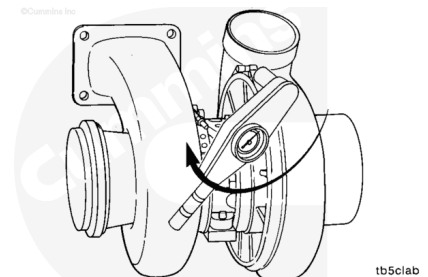


To adjust the turbocharger compressor housing alignment:

- Loosen the V-band clamp.
- Turn the compressor housing to the correct alignment.
- Tighten the clamp.

Torque Value: 9 n•m [80 in-lb]

- Tap the clamp with a mallet. Tighten the clamp again.



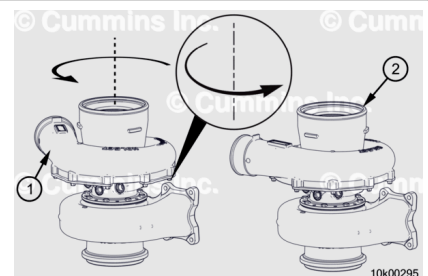
High-Pressure Turbocharger:

⚠ CAUTION ⚠

Do not lift compressor housing. Axial movement of compressor housing can result in component damage.

⚠ CAUTION ⚠

Removing more than one clamp plate at a time can result in component damage.



The HE850 turbocharger uses a bolted compressor housing joint.

If compressor housing orientation with respect to diffuser does **not** match turbocharger being replaced (2), place turbocharger on turbine housing outlet flange on clean sturdy work bench.

Secure the turbocharger and loosen compressor housing to diffuser mounting capscrews and clamp plates until housing rotates freely.

Do **not** lift compressor housing.

Rotate compressor housing to match orientation of turbocharger being replaced (1).

Tighten the capscrews in the order shown.

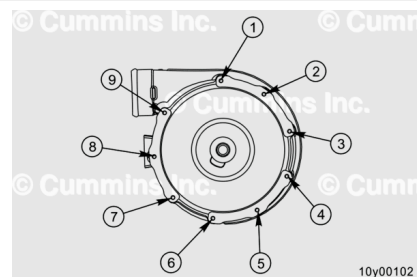
Compressor housing bolts shall be double tightened. Tighten bolts once in the circular sequence shown and repeat tightening each bolt again to the same value in the same circular sequence.

Torque Value:

M8 Bolts 33 n•m [24 ft-lb]

Torque Value:

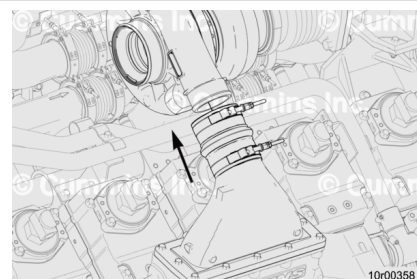
M10 Bolts 50 n•m [37 ft-lb]



Install the hose and two hose clamps between each turbocharger and aftercooler inlet connection.

Tighten the clamps.

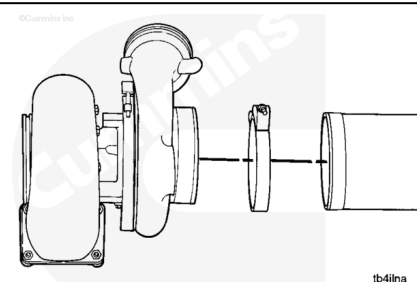
Torque Value: 9 n•m [80 in-lb]



Connect the inlet piping to the turbocharger.

Tighten the clamps.

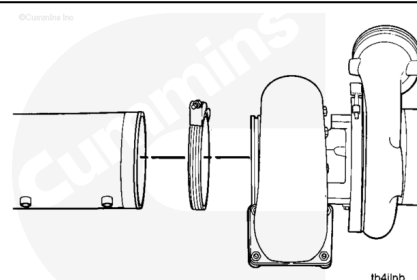
Torque Value: 9 n•m [80 in-lb]



Connect the exhaust piping to the turbocharger.

Tighten the V-band clamp.

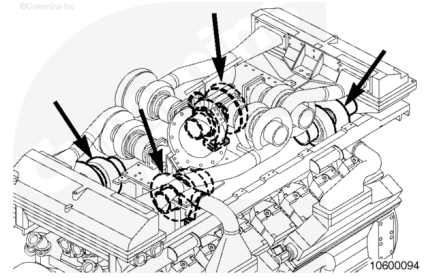
Torque Value: 9 n•m [80 in-lb]



QSK60 Two Stage Industrial and Power Generation

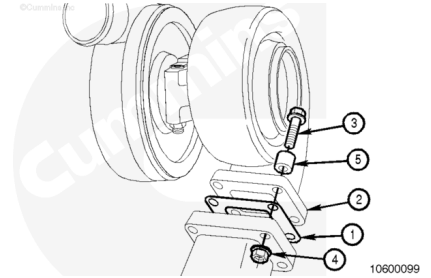
Determine which of the four high-pressure turbochargers requires installation:

- Left bank rear
- Right bank front
- Left bank front
- Right bank rear.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Note : Make sure that the turbocharger is correctly aligned during installation to prevent damage to the turbocharger bellows

Apply an anti-seize compound to the mounting capscrews threads.

The word OUT or the raised gasket bead **must** be positioned toward the turbocharger.

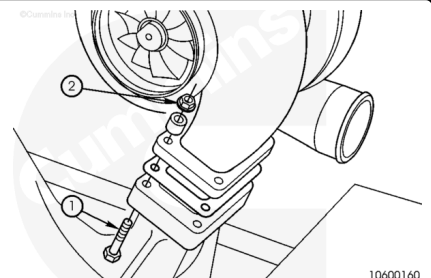
Install the gasket.

Do **not** install the turbocharger without spacers.

Install the turbocharger (2) and gasket (1) with four capscrews (3), spacers (5), and flange nuts (4).

⚠ CAUTION ⚠

Do not reuse old capscrews (1) or flange nuts (2). Discard these components. Reusing these components can contribute to exhaust leaks.

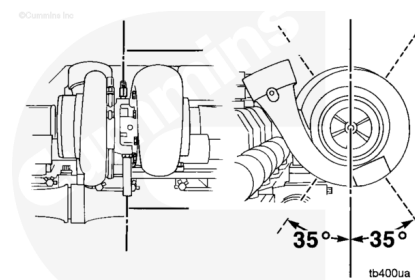


Tighten the flange nuts (4).

Torque Value: 55 n·m [40 ft·lb]

Position the turbocharger drain tube. The drain tube **must** be within 35 degrees of vertical. Turn the bearing housing to align the tube, if necessary.

The turbocharger uses capscrews to attach the bearing housing to the turbine housing.

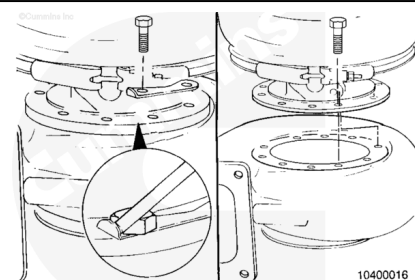


Bearing Housing Adjustment

Bend the lockplate off the capscrew heads.

Remove the capscrews.

Rotate the bearing housing. Align the drain tube and the capscrew holes.

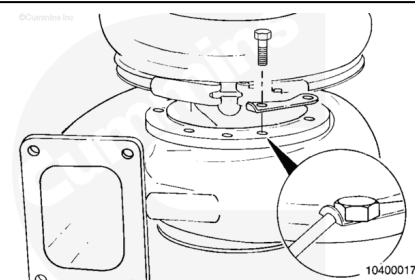


Install the lockplates and capscrews.

Tighten the capscrews.

Torque Value: 21 N·m [186 in-lb]

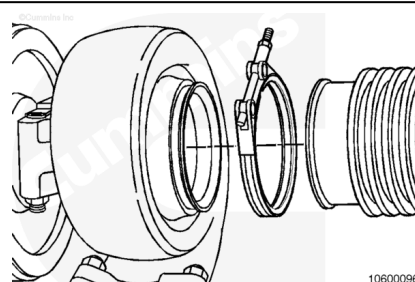
Bend the lockplate tabs over the capscrews.



Install the V-band clamp between the bellows and turbine side of the turbocharger.

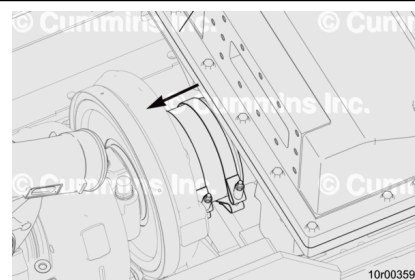
Tighten the V-band clamp.

Torque Value: 9 N·m [80 in-lb]



Install the hose and two hose clamps between the turbocharger inlet and the intercooler.

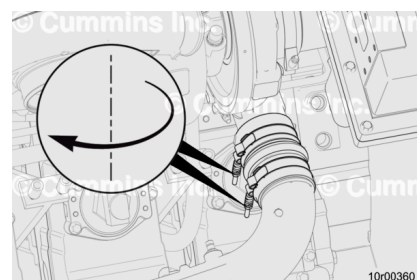
Slide the hose toward the turbocharger.



Install the hose and two hose clamps between each turbocharger and aftercooler inlet connection.

Tighten the clamps.

Torque Value: 9 n•m [80 in-lb]

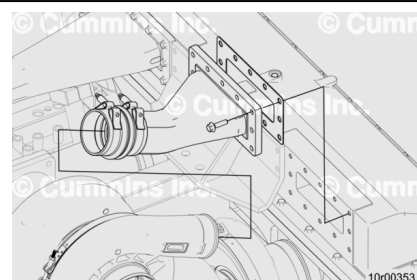


Left Bank Rear or Right Bank Front

Connect the intercooler intake air crossover, located between the low-pressure turbocharger and the intercooler, to the intercooler.

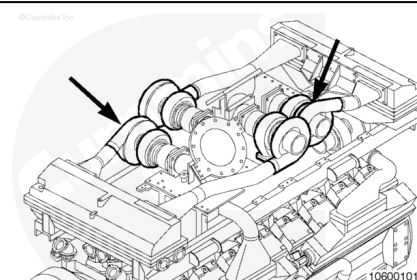
Tighten the 12 capscrews.

Torque Value: 45 n•m [33 ft-lb]



The left bank rear and right bank front turbochargers are located under the low-pressure turbochargers.

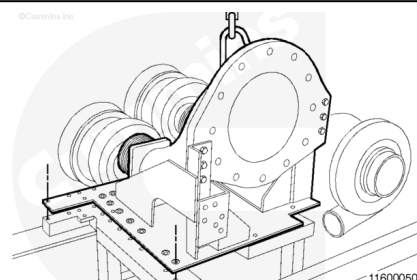
Install the low-pressure turbocharger directly above the high-pressure turbocharger being installed. Refer to Procedure 010-035 in Section 10. (/qs3/pubsys2/xml/en/procedures/56/56-010-035-tr.html)



Left Bank Front or Right Bank Rear

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



⚠ CAUTION ⚠

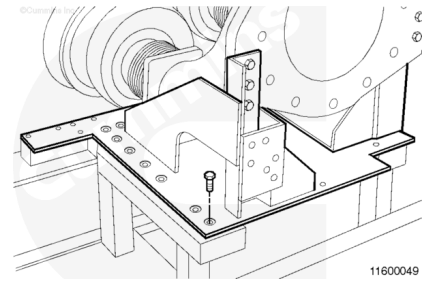
Care must be taken not to damage the bellows mounted to the exhaust outlet flange while setting the mounting plate in place.

Position the exhaust outlet flange and mounting base on the two-stage frame.

Install the thirty capscrews connecting the exhaust outlet mounting base to the two-stage frame.

Tighten the thirty mounting capscrews.

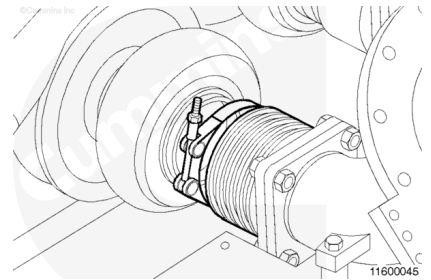
Torque Value: 45 n•m [33 ft-lb]



Connect the bellows on each leg of the exhaust outlet flange to the low-pressure turbocharger outlet.

Tighten the V-band clamps.

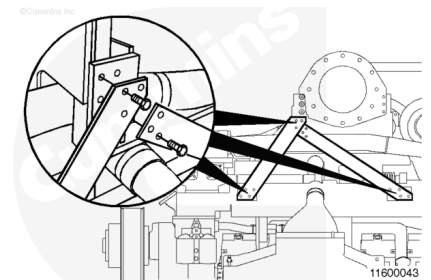
Torque Value: 8 n•m [71 in-lb]



Install the two support braces between the two-stage frame and the exhaust outlet connection mounting base.

Tighten the seven capscrews on each support brace.

Torque Value: 80 n•m [59 ft-lb]



Install the top cross support brace (1) between the two exhaust outlet connections (2).

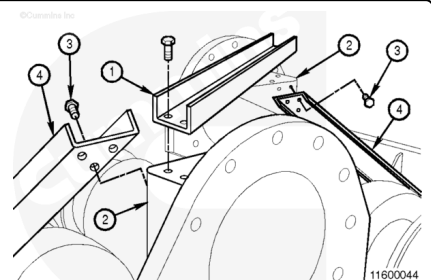
Tighten the eight capscrews (3).

Torque Value: 80 n•m [59 ft-lb]

Install the two inboard support struts (4).

Tighten the three capscrews (3) at each end.

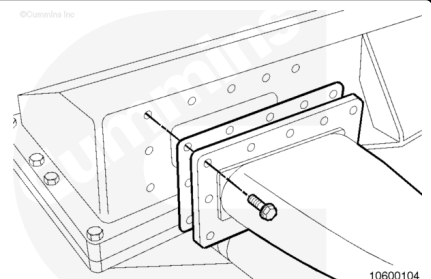
Torque Value: 45 n•m [33 ft-lb]



Connect the intercooler intake air crossover, located between the low-pressure turbocharger and the intercooler, to the intercooler.

Tighten the twelve capscrews.

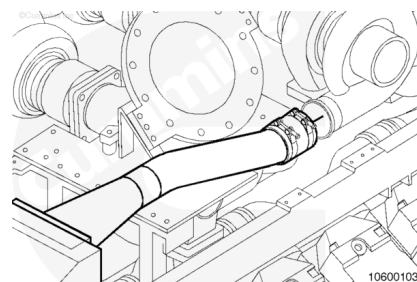
Torque Value: 45 n•m [33 ft-lb]



Install the hose clamp connecting the intercooler intake air crossover and the low-pressure turbocharger.

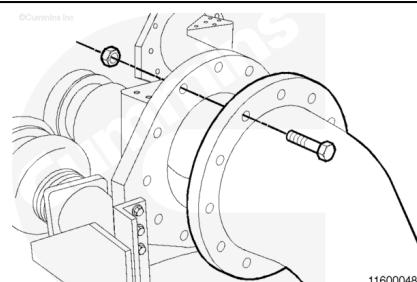
Tighten the hose clamps.

Torque Value: 8 n•m [71 in-lb]



Install the exhaust piping on the exhaust outlet flange with twelve capscrews and nuts.

Tighten the capscrews.



Finishing Steps

Industrial and Power Generation

- Install the required exhaust connection pipework. Refer to Procedure 011-001 in Section 11.
(/qs3/pubsys2/xml/en/procedures/56/56-011-001-tr.html) Refer to Procedure 011-002 in Section 11.
(/qs3/pubsys2/xml/en/procedures/56/56-011-002.html)
- Connect the turbocharger oil drain lines. Refer to Procedure 010-045 in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-045-tr.html)
- Connect the turbocharger oil supply lines. Refer to Procedure 010-046 in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-046-tr.html)
- Connect the turbocharger coolant supply and return lines. Refer to Procedure 010-041 in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-041.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section 10. (/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)
- Connect the batteries. See equipment manufacturer service information.



- Operate the engine to the required coolant temperature and check for leaks.

QSK60 Marine Applications

- Install the required exhaust connection pipework. Refer to Procedure 011-001 in Section 11.
(/qs3/pubsys2/xml/en/procedures/56/56-011-001-tr.html) Refer to Procedure 011-002 in Section 11.
(/qs3/pubsys2/xml/en/procedures/56/56-011-002.html)
- Connect the turbocharger oil drain lines. Refer to Procedure 010-045 in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-045-tr.html)
- Connect the turbocharger oil supply lines. Refer to Procedure 010-046 in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-046-tr.html)
- Connect the turbocharger coolant supply and return lines. Refer to Procedure 010-041 in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-041.html)
- Install the turbocharger inlet connections. Refer to Procedure 010-130 in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-130-tr.html)
- Install the turbocharger heat shield access panels.
- Install the aftercooler heat shields. Refer to Procedure 010-129 in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-129-tr.html).
- Install the turbocharger heat shields. Refer to Procedure 010-076 in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-076-tr.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)
- Operate the engine to the required coolant temperature. Check for leaks.

